CUSTOMER CHURN ANALYSIS

Telecom Industry | Project Completion Report

1. Introduction

Customer churn occurs when a customer stops using a telecom provider's services. In a competitive telecom market, churn can reduce recurring revenue and increase the cost of acquiring replacement customers. This project demonstrates how customer data, SQL queries, and machine learning can be combined to explore churn patterns, estimate churn risk, and support targeted retention strategies.

2. Abstract

The project develops a customer churn analysis and prediction workflow using Python, SQL, and a Streamlit dashboard. The dataset contains 1,200 synthetic customer records with fields such as tenure, contract type, internet service, payment method, monthly charges, total charges, and churn status. Data is explored and aggregated, then a logistic regression classifier is trained using a stratified train/test split. Model performance is assessed with accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. The dashboard presents summary indicators, churn patterns, sample predictions, customer segments, and example SQL results. Because the data is synthetic, findings are illustrative and must be validated on real telecom data before operational use.

3. Tools Used

- **Visual Studio Code:** code editing and project execution.
- **Python:** main programming language for data processing and modelling.
- **Pandas and NumPy:** data loading, cleaning, and analysis.
- **Scikit-learn:** preprocessing, logistic regression, and model evaluation.
- **SQLite and SQL:** storing records and performing aggregations.
- **Streamlit and Plotly:** interactive dashboard and visualizations.
- **Matplotlib / SHAP:** notebook plots and optional model explainability.

4. Steps Involved in Building the Project

1. **Dataset preparation:** load the CSV, inspect data types, check missing values and duplicates, and define churn as the target variable.
2. **SQL analysis:** load customer records into SQLite and aggregate churn by contract and payment method.
3. **Exploratory analysis:** compare churn rates across customer groups and visualize patterns by contract and tenure.
4. **Model development:** split data into training and test sets; impute missing values, scale numerical features, one-hot encode categorical features, and train logistic regression.
5. **Evaluation:** assess accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix on held-out test data.
6. **Dashboard and segmentation:** display KPIs, interactive charts, SQL examples, customer segments, and sample churn predictions.
7. **Retention recommendations:** prioritize high-risk customers, investigate service issues, test targeted offers, and monitor model quality and business impact.

5. Conclusion

The project demonstrates an end-to-end workflow for analyzing and predicting telecom customer churn with accessible, mostly free tools. Combining SQL summaries, machine learning, and an interactive dashboard makes results easier to explore and communicate. Recommended next steps are to validate the model on real, authorized telecom data, add call duration, complaint counts and recharge frequency, implement SHAP or ELI5 explanations, and test retention actions using controlled experiments. The current synthetic dataset is suitable for demonstration and learning, not for claiming real-world churn drivers or production performance.